

5-Mark Questions & Answers -- Data Structures Module 6: Sorting & Searching Techniques

Q1. Explain Bubble Sort, Insertion Sort, and Selection Sort with step-by-step trace and C programs.

Detailed algorithms, code implementations, and complexity analysis ($O(N^2)$ average, $O(1)$ space).

Q2. Explain Merge Sort and Quick Sort algorithms with Divide-and-Conquer recurrence relations.

* Merge Sort: Divide in half, recursively sort, merge halves. $T(N) = 2T(N/2) + O(N)$ implies $O(N \log N)$ in all cases.

* Qu

Q3. Explain Collision Resolution Techniques in Hashing with numerical examples.

Detailed walkthrough of Linear Probing, Quadratic Probing, Double Hashing, and Separate Chaining.